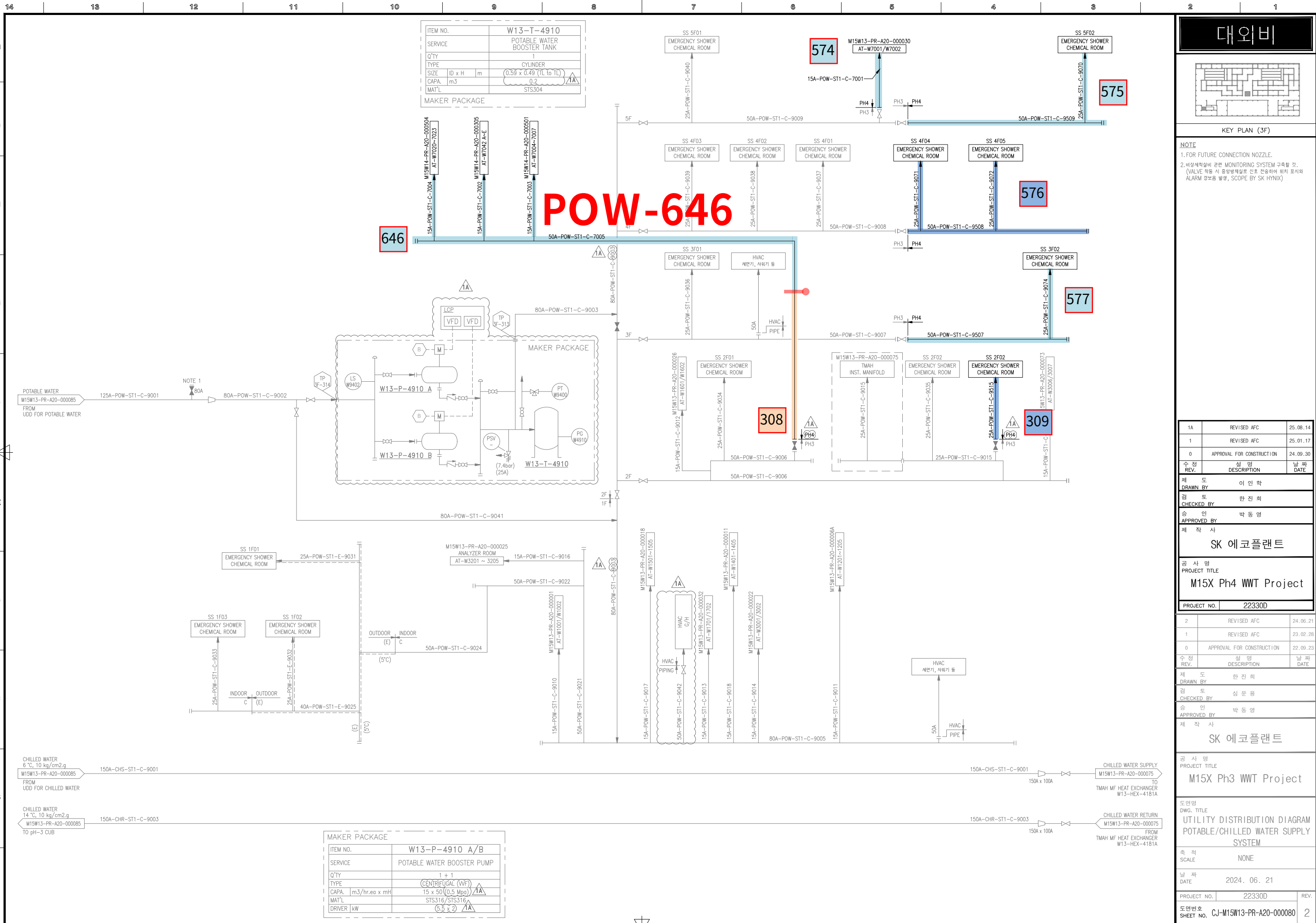


DRAWING LIST FOR TEST PACKAGE**POW-646**

PROJECT NO.: 22328D

PROJECT TITLE : M15X Ph2 Project

No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	
1	646	1FA	15A-POW-ST1-C-7002-01	1 OF 1	1	CJ-M15W13-PR-A20-000080	2	
2	646	1FA	15A-POW-ST1-C-7003-01	1 OF 1	1	CJ-M15W13-PR-A20-000080	2	
3	646	1FA	15A-POW-ST1-C-7004-01	1 OF 1	1	CJ-M15W13-PR-A20-000080	2	
4	646	1FA	50A-POW-ST1-C-7005-01	1 OF 1	0	CJ-M15W13-PR-A20-000080	2	
5	646	1MF	50A-POW-ST1-C-7005-01	1 OF 1	0	CJ-M15W13-PR-A20-000080	2	
6	646	2FA	50A-POW-ST1-C-7005-01	1 OF 2	0	CJ-M15W13-PR-A20-000080	2	
7	646	2FA	50A-POW-ST1-C-7005-02	2 OF 2	0	CJ-M15W13-PR-A20-000080	2	



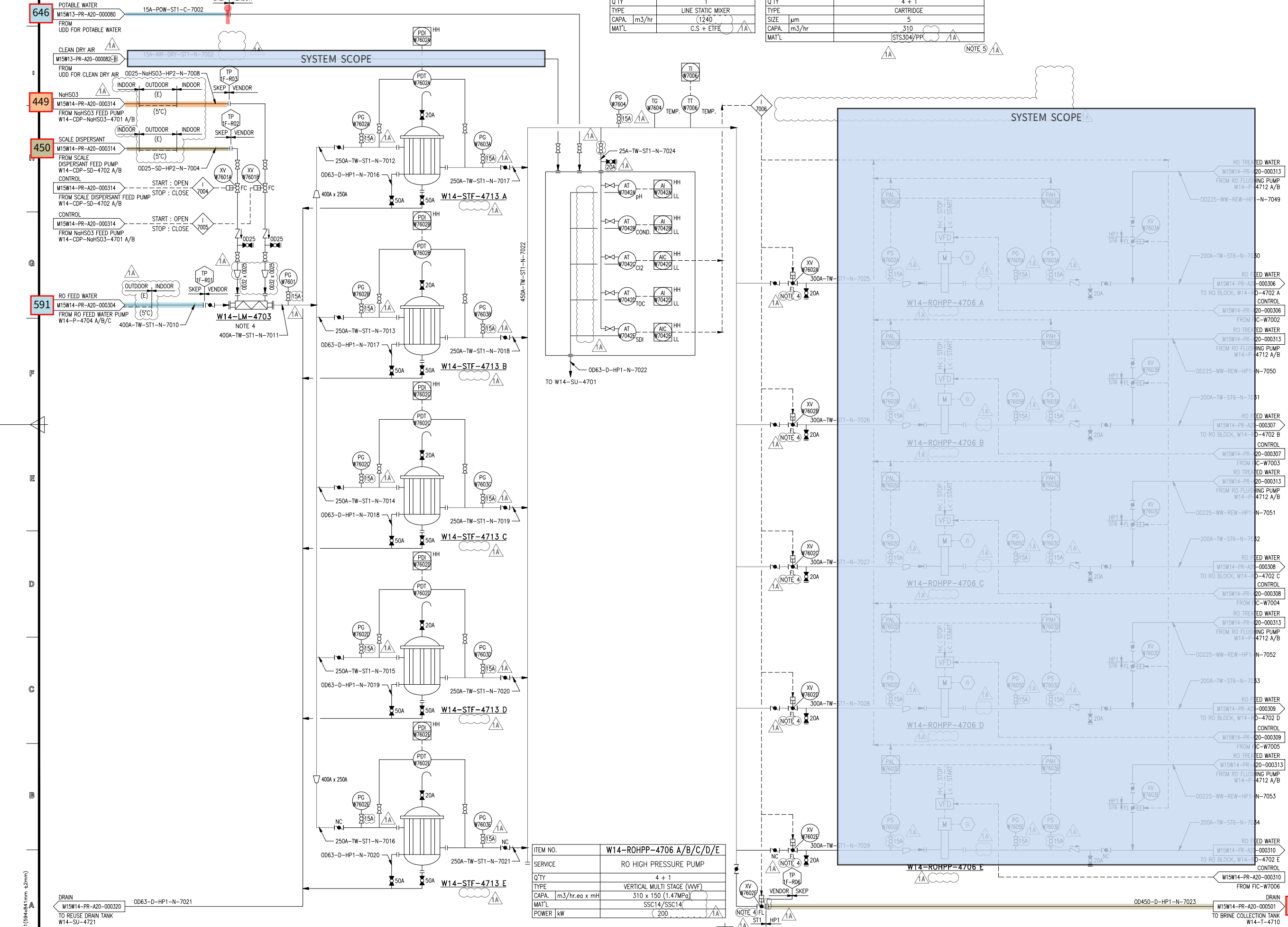
대외비

KEY PLAN (3F)

NOTE
1. FOR FUTURE CONNECTION NOZZLE.
2. 비상세척실비 관련 MONITORING SYSTEM 구축할 것.
(VALVE 작동 시 중앙방재실로 신호 전송하여 위치 표시와 ALARM 경보를 발생, SCOPE BY SK HYNIX)

1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.09.30
수정 REV.	설명 DESCRIPTION	날짜 DATE
제도 DRAWN BY	이인학	
검토 CHECKED BY	한진희	
승인 APPROVED BY	박동영	
제작사	SK 에코플랜트	
공사명 PROJECT TITLE	M15X Ph4 WWT Project	
PROJECT NO.	22330D	
2	REVISED AFC	24.06.21
1	REVISED AFC	23.02.28
0	APPROVAL FOR CONSTRUCTION	22.09.23
수정 REV.	설명 DESCRIPTION	날짜 DATE
제도 DRAWN BY	한진희	
검토 CHECKED BY	심문용	
승인 APPROVED BY	박동영	
제작사	SK 에코플랜트	
공사명 PROJECT TITLE	M15X Ph3 WWT Project	
도면명 DWG. TITLE	UTILITY DISTRIBUTION DIAGRAM POTABLE/CHILLED WATER SUPPLY SYSTEM	
축척 SCALE	NONE	
날짜 DATE	2024. 06. 21	
PROJECT NO.	22330D	REV.
도면번호 SHEET NO.	CJ-M15W13-PR-A20-000080	2

POW-646



ITEM NO.	W14-LM-4703
SERVICE	RO FEED LINE MIXER
Q'TY	1
TYPE	LINE STATIC MIXER
SIZE	1240
CAPA.	m3/hr
MAT'L	C.S + ETFE

ITEM NO.	W14-STF-4713 A/B/C/D/E
SERVICE	RO SAFETY FILTER
Q'TY	4 + 1
TYPE	CARTRIDGE
SIZE	5
CAPA.	310
MAT'L	STS304/PP

ITEM NO.	W14-ROHPP-4706 A/B/C/D/E
SERVICE	RO HIGH PRESSURE PUMP
Q'TY	4 + 1
TYPE	VERTICAL MULTI STAGE (VWF)
CAPA.	310 x 150 (1.47MPa)
MAT'L	SSC14/SSC14
POWER	200

대외비

KEY PLAN (1F)

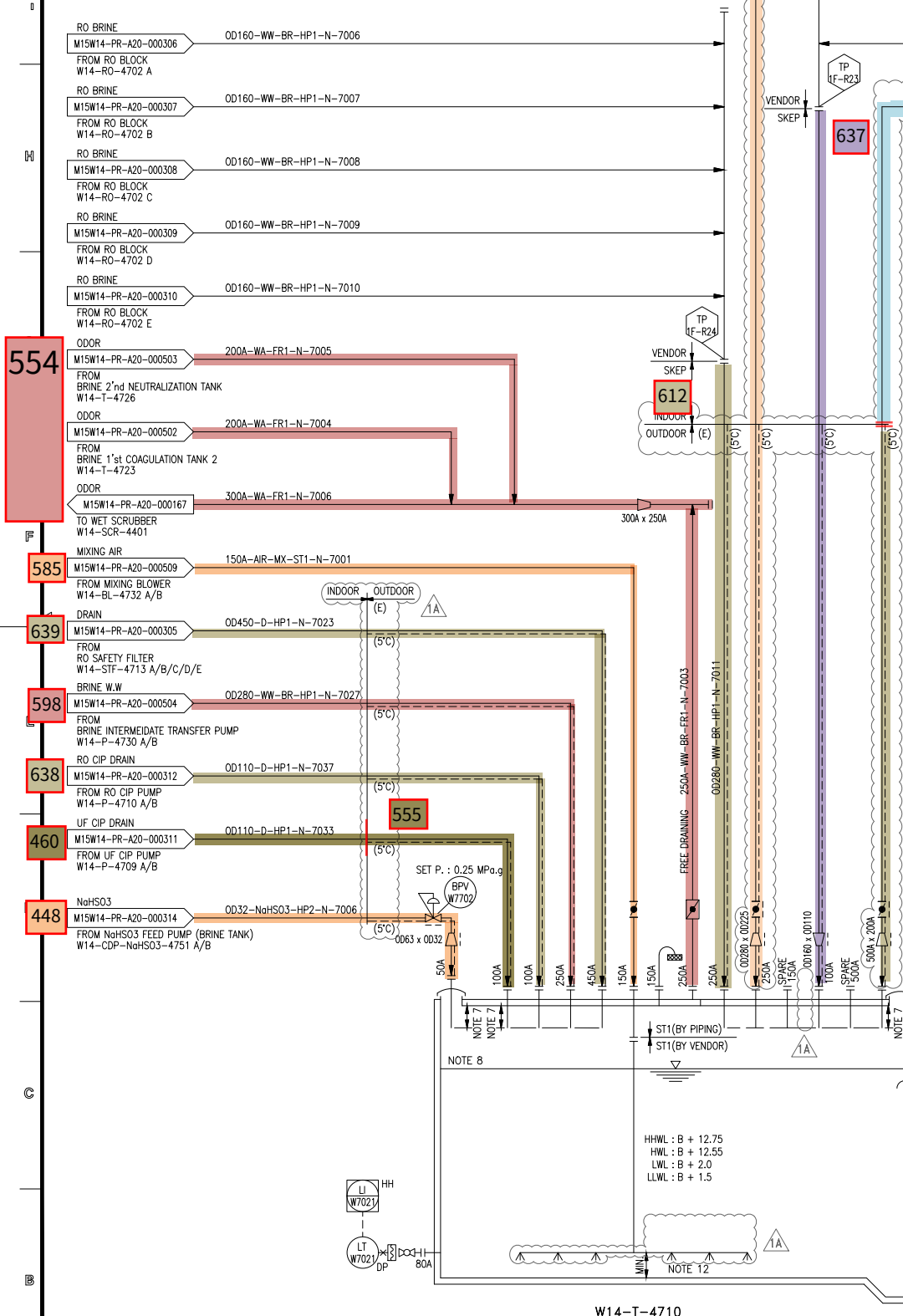
- NOTE
1. 2m 이상 높은 곳에 설치된 밸브는 CHAIN VALVE 또는 PLATFORM 구성할 것.
 2. 온천시 CLOSE 되는 배관에는 LOTO 설치.
 3. CENTRIFUGAL PUMP MAT'L 표기 : CASING / IMPELLER
 4. 50A 이상 모든 XV SPED CONTROLLER (ACCESSORY) 설치.
 5. FILTER ELEMENT 수명 : 110EA/Train

1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.10.31

수정	REV.	설명	DESCRIPTION	날짜	DATE
제	도	이	인	학	
검	토	한	진	회	
승	인	박	동	영	

SK 에코플랜트			
공 사 명 PROJECT TITLE M15X Ph4 WWT Project			
도면명 DWG. TITLE PIPING & INSTRUMENT DIAGRAM SAFETY FILTER & HIGH PRESSURE PUMP			
축 척 SCALE		NONE	
날 짜 DATE		2025. 08. 14	
프로젝트 번호 PROJECT NO.		22330D	REV. 1A
시트 번호 SHEET NO.		CJ-M15W14-PR-A20-000305	

ITEM NO.	W14-T-4710
SERVICE	BRINE COLLECTION TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H m (14.5) x 5.775 x 7.05 (EH 7.05) 5.0 x 5.3 x (14.05 (EH 12.55))
CAPA.	m3 (963 (EH 923))
HRT	hr 2.5
MAT'L	P.C + FRP



W14-T-4710

ITEM NO.	W14-P-4715 A/B/C
SERVICE	BRINE TRANSFER PUMP
Q'TY	2 + 1
TYPE	CENTRIFUGAL (WVF)
CAPA.	m3/hr.ea x mH 250 x 30 (0.29MPa)
MAT'L	ALLOY20/ALLOY20
POWER kW	(37)

대외비

KEY PLAN (1F)

NOTE
1. TANK의 HWL 이하로 연결되는 모든 배관은 BLOCK 별본 설치할 것
2. 환수 인입지점은 환수 출입측에서 최대한 이격하여 설치
3. 2m 이상 높은 곳에 설치된 별본은 CHAIN VALVE 또는 PLATFORM 구성할 것
4. MANHOLE : 2개 이상 설치한다.
MANHOLE COVER : FRP
5. 환전시 CLOSE 되는 배관에는 LOTO 설치
6. CENTRIFUGAL PUMP
MAT'L 표기 : CASING / IMPELLER
7. TANK 내 유입, 유출 확인이 가능하고, 비산되지 않도록 최소 길이로 반영할 것 (SLAB로부터 100mm 떨어진 것)
8. 내부 FRP LINING TANK의 SLEEVE는 SUS304 + FRP LINING 적용
9. FIC-W7011/W7012 중 선택은한다.
10. INTERLOCK NO. 1-7030/7031/7032 (W14-P-4715 A/B/C AUTO STOP)
: REMOTE STATUS에서 W14-T-4710 LEVEL (LI-W7022) LOW LOW STOP 또는 W14-P-4715 A/B DISCHARGE FLOW (FAL-W7011/W7012) LOW STOP
11. BRINE TRANSFER PUMP RUN 후 30초 후 부터 동작한다.
12. (DIFFUSER HOLE 및 배관배치의 관련된 상세 설계는 VENDOR가 수행한다.)

REVISED AFC	25.08.14			
REVISED AFC	25.01.17			
APPROVAL FOR CONSTRUCTION	24.10.31			
수정	실명	날짜		
REV.	DESCRIPTION	DATE		
제	도	인	학	
DRAWN BY	이인학			
검	토	한	진	희
CHECKED BY	한진희			
승	인	박	동	영
APPROVED BY	박동영			
제	작	사		
SK 에코플랜트				
공 사 명				
PROJECT TITLE				
M15X Ph4 WWT Project				
도면명				
DWG. TITLE				
PIPING & INSTRUMENT DIAGRAM BRINE COLLECTION TANK & TRANSFER PUMP				
속				
척				
SCALE				
NONE				
날				
짜				
DATE				
2025. 08. 14				
PROJECT NO.				
22330D				
REV.				
A				
도면번호				
SHEET NO.				
CJ-M15W14-PR-A20-000501				
1A				

POW-646

밸브삭제

Reuse Flange 추가

ITEM NO.	W14-T-4730	
SERVICE	BRINE INTERMEDIATE TANK	
Q'TY	1	
TYPE	RECTANGULAR	
SIZE	W x L x H m	8.1 x 3.075 x 7.45 (EH 5.5)
CAPA.	m3	(185.6) (EH 137)
HRT	hr	0.7
MAT'L	P.C + FRP	



대외비

KEY PLAN (1F)

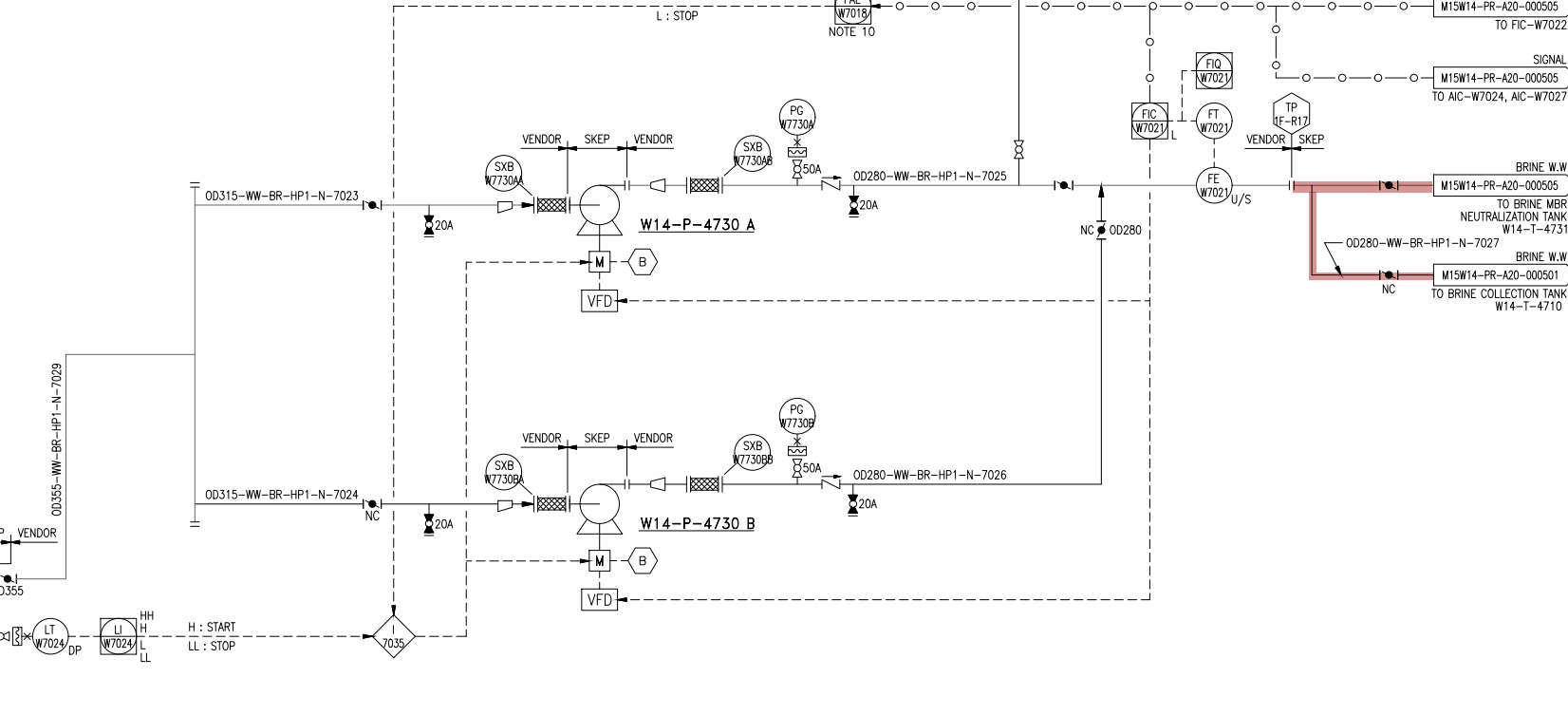
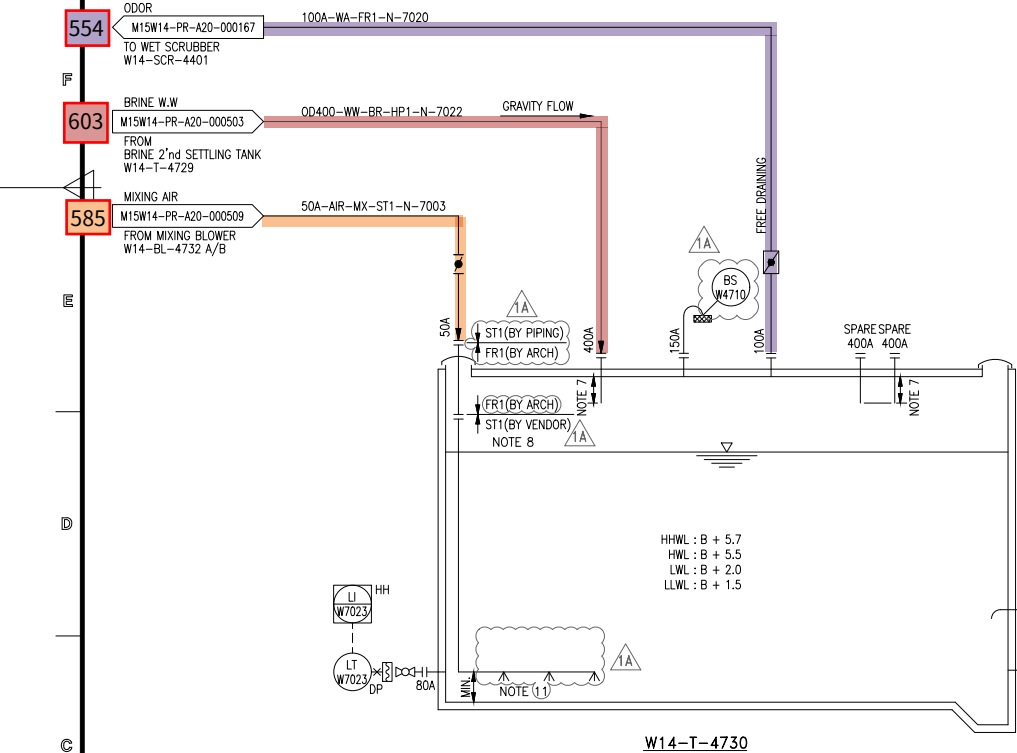
- NOTE
1. TANK와 HWL 이하로 연결되는 모든 배관은 BLOCK 별본 실시할 것
 2. 환수 인입지점은 펌프 흡입측에서 최대한 이격하여 설치
 3. 2m 이상 높은 곳에 설치된 별본는 CHAIN VALVE 또는 PLATFORM 구성할 것
 4. MANHOLE : 2개 이상 설치한다.
MANHOLE COVER : FRP
 5. 본전시 CLOSE 되는 배관에는 LOTO 설치
 6. CENTRIFUGAL PUMP
MAT'L 표기 : CASING / IMPELLER
 7. TANK 내 유입, 유출, 유입이 가능하고, 비산되지 않도록 최소 길이로 반영할 것 (SLAB 로 부터 100mm 떨어질 것)
 8. 내부 FRP LINING TANK의 SLEEVE는 SUS304 + FRP LINING 적용.
 9. INTERLOCK NO. I-7035
(W14-P-4730 A/B AUTO STOP)
: REMOTE STATUS에 W14-T-4730 LEVEL (LI-W7024) LOW LOW STOP 또는 W14-P-4730 A/B DISCHARGE FLOW (FAL-W7021) LOW STOP.
 10. BRINE INTERMEDIATE TRANSFER PUMP RUN 후 30초 후 부터 동작한다.
 11. (DIFFUSER HOLE 및 배관배치의 관련된 상세 설계는 VENDOR가 수행한다.)

POW-646

646

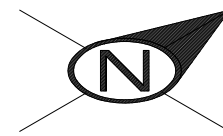
DEL

598



ITEM NO.	W14-P-4730 A/B
SERVICE	BRINE INTERMEDIATE TRANSFER PUMP
Q'TY	1 + 1
TYPE	CENTRIFUGAL (VVVF)
CAPA.	m3/hr.ea x mH
MAT'L	SSC13/SSC13
POWER kw	{37} /1A

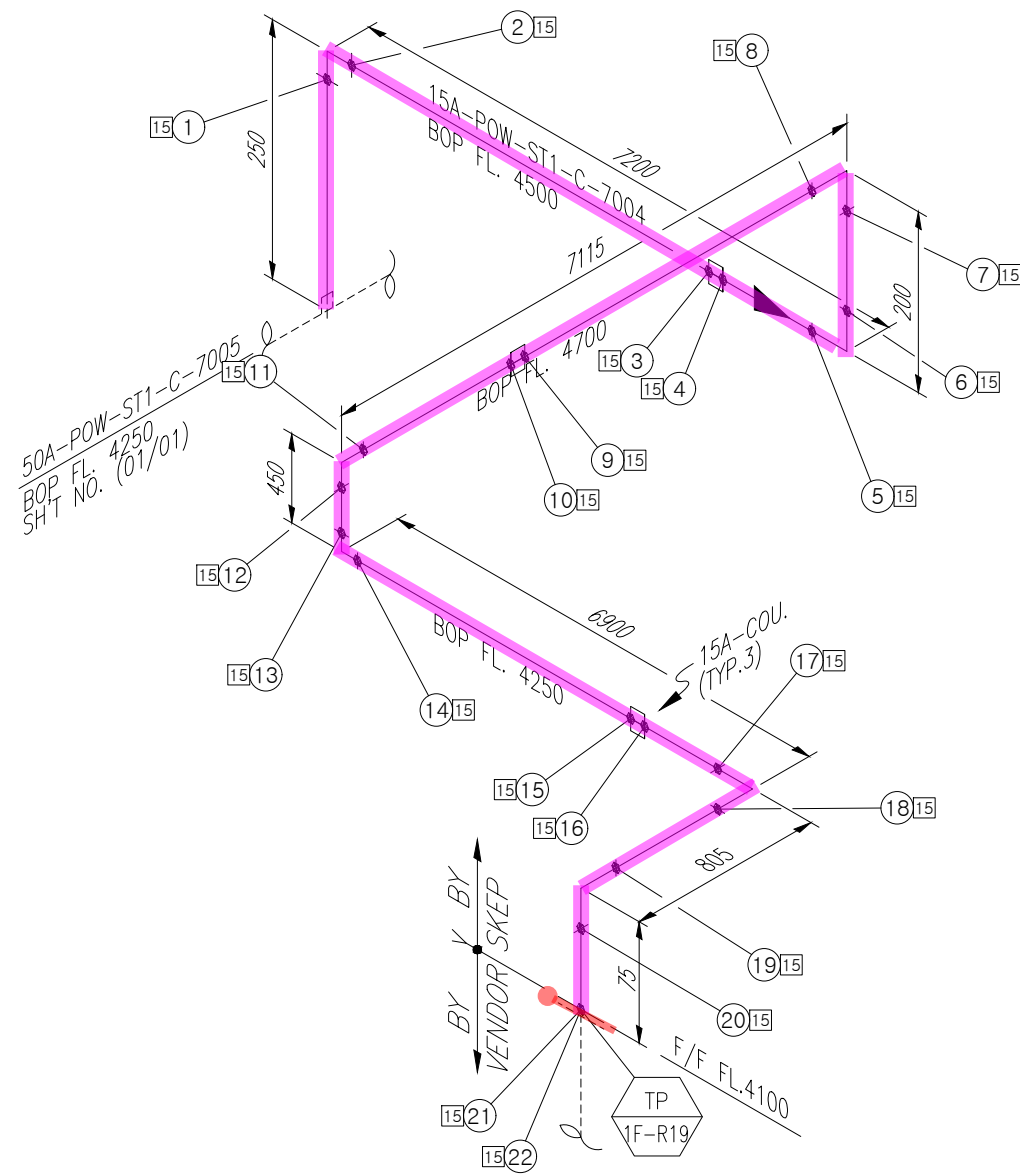
1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.10.31
수정 REV.	설명 DESCRIPTION	날짜 DATE
제 도 DRAWN BY	이 인 학	
검 토 CHECKED BY	한 진 희	
승 인 APPROVED BY	박 동 명	
제 작 사	SK 에코플랜트	
공 사 명 PROJECT TITLE	M15X Ph4 WWT Project	
도면명 DWG. TITLE	PIPING & INSTRUMENT DIAGRAM BRINE INTERMEDIATE TANK & TRANSFER PUMP	
속 척 SCALE	NONE	
날 짜 DATE	2025. 08. 14	
PROJECT NO.	22330D	REV.
도면번호 SHEET NO.	CJ-M15W14-PR-A20-000504	1A



1. (*) FIELD ADJUST

	CONTRACTOR		PROJECT NAME	
			M15X Ph4 Project	
02	COMMODITY	POW	JOB NO.	
02			22330D	
01	LINE NO.	15A-POW-ST1-C-7002	SH'T NO. 01 / 01	REV. 1

INSULATION	CODE	C	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14						
	THICK.	25	DESIGN	13	60	UNIT R1FA						
			OPERATING	10	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-010017-B		25.03.31	REVISED AS MARKED	황래중	안남후	박동
						REF. P & ID CJ-M15W13-PR-A20-000080		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동
							REV. NO.	DATE	DESCRIPTION	BY	CHK	APP

[illegible]

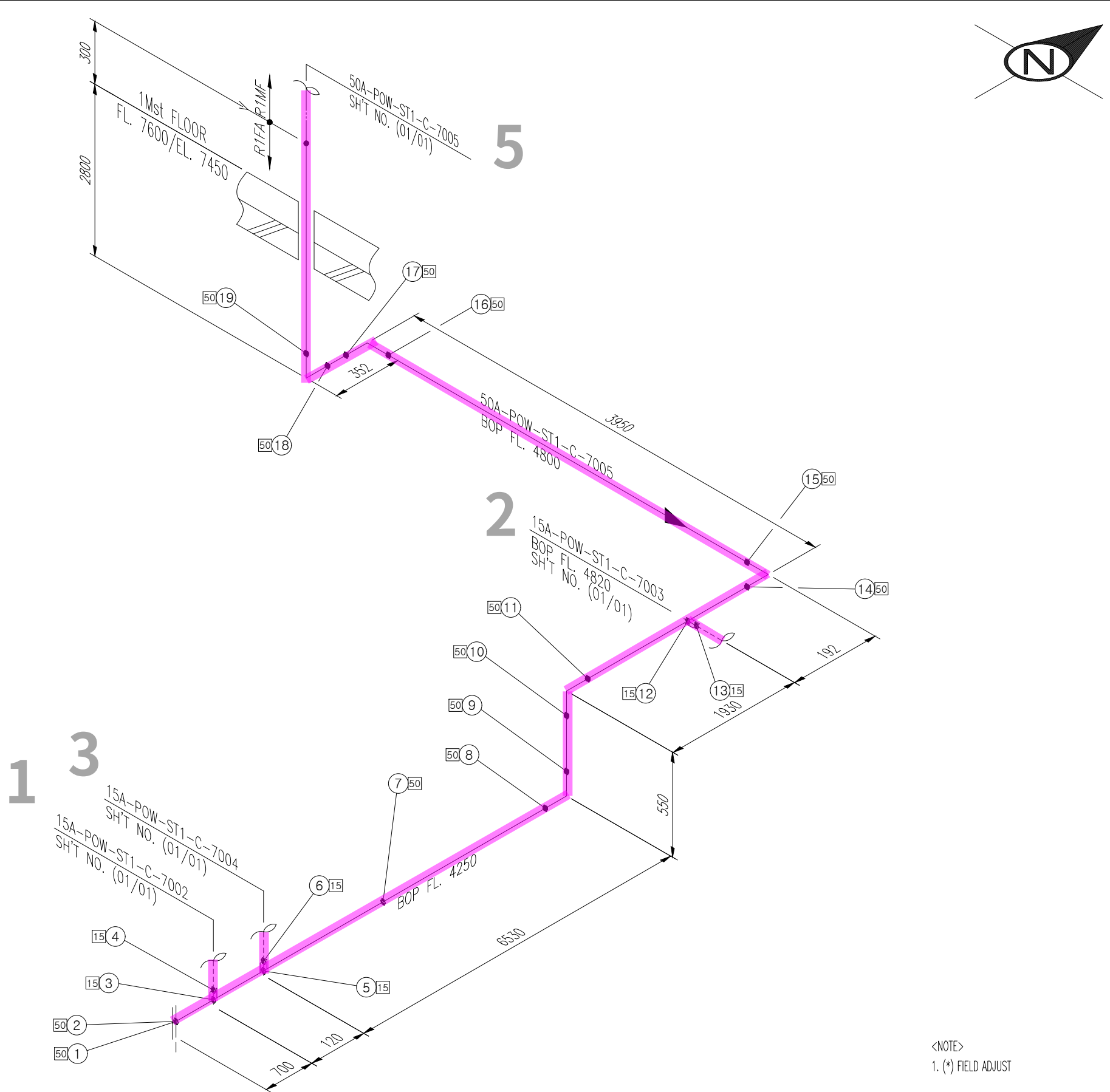
1. (*) FIELD ADJUST

CLIENT  3



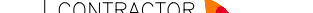
3

CONTRACTOR		PROJECT NAME	
		M15X Ph4 Project	
COMMODITY POW		JOB NO.	
		22330D	
LINE NO.	15A-POW-ST1-C-7004	SH'T NO.	REV.
		01 / 01	1



<NOTE>
1. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	C	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14							CONTRACTOR 	PROJECT NAME		
	THICK.	25	DESIGN	13	60	UNIT	R1FA								M15X Ph4 Project		
			OPERATING	10	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-010016-A		△	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	COMMODITY	POW	JOB NO.	
						REF. P & ID CJ-M15W13-PR-A20-000080								22330D			
								REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	50A-POW-ST1-C-7005	SH'T NO. 01 / 01	REV. 0

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



ITEM	SIZE	Q'TY	DESCRIPTION
ST1			
P	50A	17.5M	PIPE, STS 304TP, ERW, BE - SCH 10S
E90	50A	5EA	ELBOW 90 DEG LR, STS304, ERW, BE - SCH 10S
HF	15A	3EA	COUPLING HALF, STSF304, SW - SCH 10S
F	50A	1EA	FLG SO, KS 10K, RF, STS304
FB	50A	1EA	FLG BLIND, KS 10K, RF, STS304
G	50A	1EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	50A	1ST	MACHINE BOLT & NUTS :STS 304 / W WASHERS
IN	25tX50A	17.5M	관 보존재(고무발포)



CLIENT		CONTRACTOR SK ecoplant		PROJECT NAME M15X Ph4 Project	
		COMMODITY POW		JOB NO. 22330D	
		LINE NO. 50A-POW-ST1-C-7005		SH'T NO. 01 / 01	REV. 0

[illegible]

5

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VBLO/VB1/VB1O/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	C	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14						CONTRACTOR 	PROJECT NAME		
	THICK.	25	DESIGN	13	60	UNIT R1MF							M15X Ph4 Project		
			OPERATING	10	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-010017-D		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	COMMODITY	POW	JOB NO.
					REF. P & ID CJ-M15W13-PR-A20-000080	22330D									
						REV. NO.							DATE	DESCRIPTION	BY

